

Building a Walk-Forward Backtest with No Look-Ahead Bias and a True Out-of-Sample Test

Strategy research · L/S Gamma Desk, QUANTT

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Question: look into making a backtest that has no look ahead bias and uses walk forward optimization and out of sample testing

Method: Searched all 35 papers in `docs/papers/` (page-tagged index) + web search. Quotes verified against the cited PDF pages. Page numbers are PDF pages; printed page numbers are given where they differ.

Related research: Clustering for algo selection · Position size and exit rules · Signal generation

1 Short answer

A trustworthy backtest needs three separate protections.

1. **No look-ahead.** Every decision on day t may use only what was knowable at that decision time. Forecast models are refit on data up to t , and trades fill at prices from *after* the decision.
2. **Walk-forward optimization.** Parameters such as the VRP bands, DTE and stops are chosen on a training window and then run, unchanged, on the next unseen window. Repeat, rolling forward. Leave a gap (an **embargo**) at least as long as a trade's holding period between training and test, so trades don't straddle the boundary.
3. **A final hold-out.** Lock away a last slice of data, e.g. the final 12 months, and run the chosen configuration on it **exactly once**. Every look at it after that makes it in-sample.

Two more things the papers stress:

- **Count every configuration you try**, and judge the winner with a selection-aware statistic, such as the Deflated Sharpe Ratio or the probability of backtest overfitting.
- **Use realistic option fills**, not the mid price.

The simplest way to stay honest on this desk is to make the backtest call **the same signal, selection, exit and P&L code as the live runner**, fed day by day.

2 What the papers say

2.1 Walk-forward: refit on the past, test on the next unseen period

- **How it works**, as used in a repo paper on real S&P 500 options:

“In each iteration, the model is re-estimated using the most recent information, and performance is assessed on the next unseen period.” (Bracha, Sakowski & Michaków 2025, *DRL for ATM S&P 500 Options Hedging*, p. 20, printed p. 19)

- **Tuning happens only inside the training data:**

“Hyperparameters are tuned exclusively on the training and validation time periods to avoid information leakage, with the validation year performance serving as the criterion for selecting the best configuration.” (same, p. 21, printed p. 20)

- **The window lengths are themselves a choice that changes results:**

“the performance of a trading strategy using the Exponential Moving Average (EMA) evaluated within a walk-forward procedure based on the Robust Sharpe Ratio is highly dependent on the chosen window size.” (Mroziewicz & Ślepaczuk 2026, *Double Out-of-Sample Walk-Forward Optimization*, p. 1)

So pick the window lengths in advance, or treat them as one more parameter chosen inside training.

- **Retraining is part of the method**, not an afterthought:

“Generalization relies on walk-forward discipline and policies should be retrained periodically as volatility regimes evolve.” (*Deep Hedging with RL: A Practical Framework*, p. 8)

2.2 A final hold-out, used once

- **Mroziewicz & Ślepaczuk split the data into two periods first**, then ran walk-forward only on the first:

“All strategy parameter optimization was performed on the Global Training Data Period, while the Unseen Data Period was used only once to calculate the performance for the final best parameter combination.” (Mroziewicz & Ślepaczuk 2026, p. 7)

- **Wysocki does the same on S&P 500 options**, with a four-window walk-forward and then:

“a strictly held-out 2025 out-of-time slice that is never used during training, hyperparameter search, or model selection.” (Wysocki 2026, *Harvesting the VRP: A Learning-to-Rank Approach*, p. 3)

2.3 Sources of look-ahead to guard against

- **Timing of returns versus decisions:**

“The simulator eliminates look-ahead bias by aligning returns with the action timestamp” (*Deep Hedging with RL: A Practical Framework*, p. 2)

- **Model and feature choices made on the full sample.** Wysocki reruns feature selection inside each window:

“Running the pipeline per window prevents look-ahead leakage from later periods into earlier-window selections” (Wysocki 2026, p. 9)

- **Regime models that peek ahead.** A regime model fitted over the whole history labels day t using later days. Shu, Yu & Mulvey instead infer each day’s regime online, saying they

“need to infer the state to which day t belongs based on the features” available at the end of that day, “without access to future data” (Shu, Yu & Mulvey 2024, *Statistical Jump Model Approach*, p. 11)

- **Execution delay belongs in the validation too.** Shu et al. tune with the same trading delay as the test:

“we align the validation period with the out-of-sample testing period by assuming the same delay” (same, p. 16)

- **Using true future values of features in multi-step forecasts:**

“using feature’s real values introduces look-ahead bias as the model forecasts future days based on previous predictions.” (Blake, Gandhi & Jakkula 2025, *Improving S&P 500 Volatility Forecasting through Regime-Switching*, p. 12)

2.4 Leakage across the train/test boundary, and too many tries

- **An embargo gap between training and validation:**

“Embargo period: 30 days between train and validation to prevent data leakage” (Sheppert 2026, *The GT-Score*, p. 3)

- **Multiple testing:**

“When many trading rules or parameterizations are tried, the best in-sample performer is likely to be a false discovery unless multiple testing is accounted for.” (Sheppert 2026, p. 2)

The same paper points to the Deflated Sharpe Ratio and the probability of backtest overfitting (PBO) as corrections.

2.5 Web: purging, embargo, selection-aware statistics, option fills

These sources are outside the repo.

- **Purging and embargo** come from López de Prado’s *Advances in Financial Machine Learning* (2018). Purging removes training observations whose outcome window overlaps the test period. An embargo then drops a short stretch after each test period from training. Combinatorial purged cross-validation (CPCV) builds many train/test splits under these rules, rather than walk-forward’s single path (overview).
- **The Deflated Sharpe Ratio** (Bailey & López de Prado 2014, *Journal of Portfolio Management* 40(5)) adjusts a Sharpe ratio for the number of trials, short samples and non-normal returns (JPM).
- **The probability of backtest overfitting** (Bailey, Borwein, López de Prado & Zhu, *J. Computational Finance* 20(4), 2017). Its abstract notes that standard hold-out methods “tend to be unreliable and inaccurate in the context of investment backtests” (Risk.net).
- **Option fills.** ORATS, a commercial options data vendor, simulates buys and sells at “the mid-point plus or minus slippage” and marks positions at the mid. It snapshots quotes “14-minutes before the close” because “market makers will often widen out as the close approaches” (ORATS).

3 How the papers connect

Source	Builds on / agrees with	Differs from	Key contribution
Bracha et al. 2025 (repo)	Baranochnikov & Ślepaczuk	Rolls the training window forward	Walk-forward with per-window tuning on S&P 500 options
Mroziewicz & Ślepaczuk 2026 (repo)	Pardo; Bracha	Treats window lengths as parameters	Walk-forward plus a single-use unseen period
Wysocki 2026 (repo)	López de Prado	Expanding windows	Per-window feature selection; strict out-of-time hold-out on S&P options
Sheppert 2026, GT-Score (repo)	López de Prado; Bailey	Changes the objective function	Embargo; multiple-testing awareness
Practical Deep Hedging (repo)	Buehler	—	Align returns to the action timestamp
Shu, Yu & Mulvey 2024 (repo)	Nystrup	Validation tuned on strategy P&L	Online inference; same delay in validation and test
López de Prado 2018 (web)	—	Combinatorial splits instead of one path	Purging, embargo, CPCV
Bailey et al. 2014, 2017 (web)	—	Statistics, not a split scheme	Deflated Sharpe ratio; PBO

- **Everyone agrees on the core:** tune only on the past, test on unseen data, and keep a final hold-out you don't reuse.
- **They differ on windows.** Bracha rolls a fixed window forward, while Wysocki expands his. Mroziewicz shows the choice matters.
- **López de Prado and Bailey go further than walk-forward.** They argue one walk-forward path is still a single noisy sample, and add combinatorial splits and selection-aware statistics.

4 Relevance to the LS Gamma desk

These are suggestions. The PM sets the windows and the parameter grid.

- **One code path.** Build `src/lsgamma/backtest/` so that each simulated day calls the existing pure modules:
 - `signals.SIGNALS`, `forecasting.FORECASTERS`, `trading.selection`, `trading.exits` and `trading.pnl`
 - with a simulated broker in place of IBKR

Then the backtest tests the code that trades, and the live and backtest rules can't drift apart.

- **Decision timestamp.** The live runner decides at about 10:15 ET. In the backtest, the decision on day t uses:
 - the forecast fitted on returns up to the **close of $t-1$**
 - VIX and features up to $t-1$
 - the option chain as of the decision time

The fill price must come from **after** the decision. Today's partial yfinance bar must never enter. (This is already an open item for the live runner too.)

- **Refit every day, walk-forward.** Refit GARCH or HAR each day on data up to $t-1$, never once on the full sample.
- **Parameters to optimize, kept to a minimum:** the long and short bands, maybe the target DTE, and the stop multiples. Every extra parameter multiplies the number of trials.
- **A suggested schedule** for the R2 options data (2014-06 to 2026-07):
 - expanding training window of at least 3 years
 - 1-year test windows, stepping 1 year
 - an embargo of about 30 trading days, longer than a ~23-day trade
 - the last 12 months (about 2025-07 to 2026-07) as the single-use hold-out
- **Positions overlap the boundary.** A trade opened in training can still be open in the test window. Either close everything at each boundary, or purge and embargo by holding period.
- **Fills and marks:**
 - Enter and exit at $\text{mid} \pm \text{slippage}$ (the same slippage as `configs/paper.toml`), or at bid/ask where quotes exist.
 - Mark at mid.
 - Add share commissions for the hedge.
- **Report honestly:**
 - Log every configuration tried.
 - Report the walk-forward test P&L stitched together, the hold-out result once, and a Deflated Sharpe ratio across all trials.

5 Gaps and open questions

- **The data has trades, not quotes.** R2 holds intraday SPY option *trades* but no bid/ask quotes and no intraday SPY price. Marks would have to be built from trades near a fixed time, and the bid-ask spread modelled.
- **Few independent trades.** About 12 years of ~23-day trades gives at most around 130 non-overlapping trades, so a large parameter grid will overfit. The Deflated Sharpe ratio and PBO become important.
- **Regime shifts.** 2014–2026 includes the COVID-19 crash and the 2022 bear market. A one-year test window can be dominated by a single episode.
- **No repo paper tests a daily-hedged single-leg SPY option strategy with this protocol.** The closest are Wysocki (0DTE short puts) and Bracha (hedging, not signal trading).
- **Look-ahead in data vendors.** Index values from yfinance (e.g. VIX) are end-of-day. Using them at 10:15 on day t is look-ahead unless lagged one day.

6 Sources

Repo papers used, with pages cited:

- `Application-of-Deep-Reinforcement-Learning-to-At-the-Money-S&P-500-Options-Hedging.pdf`: Bracha, Sakowski & Michaków (pp. 20, 21)

- `A-Novel-Approach-to-Trading-Strategy-Parameter-Optimization-Using-Double-Out-of-Sample-Data-and-Walk-Forward-Techniques.pdf`: Mroziewicz & Ślepaczuk (pp. 1, 7)
- `Harvesting-the-Volatility-Risk-Premium-A-Learning-to-Rank-Approach.pdf`: Wysocki (pp. 3, 9)
- `The-GT-Score-A-Robust-Objective-Function-for-Reducing-Overfitting-in-Data-Driven-Trading-Strategies.pdf`: Sheppert (pp. 2, 3)
- `Deep-Hedging-with-Reinforcement-Learning-A-Practical-Framework-for-Option-Risk-Management.pdf` (pp. 2, 8)
- `Downside-Risk-Reduction-Using-Regime-Switching-Signals-A-Statistical-Jump-Model-Approach.pdf`: Shu, Yu & Mulvey (pp. 11, 16)
- `Improving-S&P-500-Volatility-Forecasting-through-Regime-Switching-Methods.pdf`: Blake, Gandhi & Jakkula (p. 12)

Checked, only standard in-sample/out-of-sample splits with no protocol detail:

- `Hedging-with-Linear-Regressions-and-Neural-Networks.pdf` (rolling 900/180-day windows)
- `Realised-Volatility-Forecasting-Machine-Learning-via-Financial-Word-Embedding.pdf` (rolling estimation windows; embeddings trained only on pre-sample text)
- `A-machine-learning-approach-to-volatility-forecasting.pdf`
- the IV-surface deep hedging papers
- OPHR, PRISM (simulated)

The other papers have no backtest methodology.

Web sources (outside the repo):

- Purged cross-validation overview (López de Prado 2018, *Advances in Financial Machine Learning*)
- Bailey & López de Prado 2014, *The Deflated Sharpe Ratio* (JPM 40(5))
- Bailey, Borwein, López de Prado & Zhu 2017, *The Probability of Backtest Overfitting* (J. Comp. Finance 20(4))
- ORATS, *Answers to Common Questions: Backtesting*

Suggested papers to add to docs/papers/:

1. **Bailey & López de Prado 2014, *The Deflated Sharpe Ratio***: the statistic for judging a winner chosen from many trials.
2. **Bailey et al. 2017, *The Probability of Backtest Overfitting***: measures how likely the chosen configuration is overfit.
3. **López de Prado 2018, chapters 7 and 12** (cross-validation and backtesting through CPCV): the purging and embargo method.